

LTIMB10 - 26.11.2025 - D2 - DWH Architecture & Data Modelling

1. Data Warehouse Architecture

A Data Warehouse (DWH) is built in **layers**, just like a factory assembly line.

Raw materials → processed → packaged → delivered.

Similarly, in a DWH:

Layer 1: Staging Layer (RAW Layer)

- This is where **raw data lands first**.
- No cleaning, no validation — *just raw files*.

Example:

An e-commerce company receives raw CSV/JSON files from:

- Orders system (OLTP)
- Payments API
- Website logs
- Mobile App events

All these land into **Staging Layer**, e.g., Snowflake stage, S3 bucket, or Azure ADLS.

Layer 2: Integration Layer (Data Warehouse / ODS Layer)

- This is where data is **cleaned, standardized, transformed**.
- Duplicate customers removed.
- Product names standardized.
- Data types corrected (date formats, currencies).
- Fact and Dimension tables are created here.

Example:

Product categories like:

- “Electronics”
- “electronic”
- “ELECTRONICS”

→ All standardized to “**Electronics**”.

Layer 3: Access Layer (Data Marts / Reporting Layer)

This layer is optimized for **BI dashboards** and **business users**.

Contains:

- Sales Data Mart
- Marketing Data Mart
- Inventory Data Mart

These are usually **star schema models** (facts + dimensions).

Example:

A Tableau dashboard reads data from **sales_data_mart.fact_sales**.

2. Enterprise Data Warehouse (EDW) vs Data Marts

EDW = The Big Library (All Subjects)

- Stores information for the entire company.
- Centralized.
- Integrates all departments (Sales, Finance, HR, Marketing).
- Single source of truth.

Example:

Walmart's EDW contains:

- All stores
 - All customers
 - All products
 - All sales
 - All suppliers
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Data Mart = One Section of the Library

A subset of the EDW, focused on **one business area**.

Examples:

- Sales Data Mart → KPIs for revenue, orders, customers
- HR Data Mart → employees, attendance, salaries
- Finance Data Mart → budgets, expenses, profits

Why Data Marts?

- Faster queries
 - Tailored for business teams
 - Reduced cost
 - Easier security management
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3. ETL vs ELT Processes

ETL = Extract → Transform → Load

- Data is transformed *before* loading into DWH.
- Traditional method (Informatica, SSIS, Talend).
- Used when transform engine ≠ data warehouse.

Analogy:

Washing clothes before storing them in your cupboard.

Industry example:

A bank using SSIS transforms data on-premise and loads into their Oracle DWH.

ELT = Extract → Load → Transform

- Data is loaded *first* into DWH and then transformed inside the warehouse.
- Used in cloud warehouses like Snowflake, BigQuery, Redshift.
- Faster because cloud DWHs have massive compute.

Analogy:

Putting clothes in the cupboard and arranging them later inside the cupboard.

Industry example:

Netflix loads raw event logs first, then transforms using Snowflake/Databricks.

When to use ETL vs ELT?

Scenario	Choose
On-prem databases	ETL
Cloud Data Warehouse	ELT
Heavy transformations	ELT
Sensitive data that must be cleaned before loading	ETL
Need for rapid ingestion	ELT

4. Real-time vs Batch Data Warehousing

Batch Processing (Most Common)

- Data loads happen on a schedule: hourly, daily, weekly.
- Used for dashboards and historical analytics.

Example:

Flipkart updates Sales Dashboard **every night** at 2 AM.

Pros:

- Stable
- Cost-effective
- Best for reporting

Cons:

- Not real-time
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Real-time / Near Real-time Processing

- Data updates continuously (every few seconds/minutes).
- Used for fraud detection, live inventory, recommendation engines.

Examples:

- Swiggy showing live order tracking
- Uber updating driver locations
- Banks detecting fraud transactions instantly

Tech Used:

Kafka, Snowpipe Streaming, Airbyte, Debezium, Fivetran.

When to Use What?

Requirement	Batch	Real-Time
Daily sales dashboard	✓	✗
Fraud detection	✗	✓
Inventory available now	✗	✓
Monthly financial reporting	✓	✗

5. Cloud-based Data Warehousing

Cloud-based DWH = Data Warehouse hosted on cloud.

Examples:

- Snowflake
 - Google BigQuery
 - Amazon Redshift
 - Azure Synapse
 - Databricks Lakehouse
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Why companies prefer cloud DWH?

1. No servers to manage

Cloud provider manages:

- Storage
- Compute
- Backups
- Performance

2. Scales automatically

Need more power?

Click 1 button.

3. Pay-as-you-go

Pay only for:

- Data stored
- Compute time used

4. High performance

Massive parallel processing (MPP) → super fast.

5. Integrates easily

Connect with S3, Azure, APIs, Kafka, Python, BI tools.

Real Industry Example:

Amazon uses Redshift to analyze:

- Customer purchases
- Shipping delays
- Inventory levels

Netflix uses Snowflake for:

- User behavior analytics
 - Content recommendations
 - A/B testing results
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Final Notes:

Concept	Simple Meaning
Staging Layer	Raw data landing area
Integration Layer	Cleaned, standardized warehouse tables
Access Layer	Data marts for dashboards
EDW	Central warehouse for whole company
Data Mart	Mini-warehouse for teams (sales, finance)
ETL	Transform first → then load
ELT	Load first → then transform
Batch	Scheduled loads (daily/hourly)
Real-time	Instant data processing
Cloud DWH	Snowflake/BigQuery-based warehouse
